GYANMANJARI INNOVATIVE UNIVERSITY

Gyanmanjari Institute of Technology – Faculty of Engineering &Technology

Hypothesis Formulation and Testing

Formulating hypotheses from the literature review and testing them on experimental data

Student Name	Kothari Tarj Mehulbhai	Enrollment No.	260205020001
Subject	Research Methodology (METXX11501)	Semester	1 (M.Tech)
Activity	ALA 2	Marks	10

Researchproblemusedintheseactivities: ComparativeSearchEfficiencyofaRandomizedSkipList,a DeterministicRed-BlackTreeandaSelf-AdjustingSplayTreeunderDifferentInsertionOrdersandAccess Patterns.
(Illustrativeproblemselectedforthissubmission;replaceitwiththefaculty-assignedproblemifadifferentoneis given.)

1. Aim

Based on the literature review (ALA 1), to formulate suitable research hypotheses and to apply suitable statistical techniques to **test and interpret** them. The data were produced by a replicated computer experiment (30 independent replications per condition); all test results below were computed from that data (file `rm_experiment_data.csv`).

2. Basic Concepts Applied

Concept	Meaning in this study
Nullhypothesis (H_0)	Statementofnodifference/noeffect(forexample,twostructureshavethesame mean search cost).
Alternativehypothesis (H_1)	Statementthatadifferenceexists;*two-tailed*($\neq$)or*directional*(<,>)whenthe literature predicts a direction.
Significance level (α)	0.05: the accepted probability of a Type I error (rejecting a true H_0).
p-value	Probability of obtaining a result at least as extreme asobserved if $H_0$is true. Reject $H_0$when $p<\alpha$.
Type II error (β) / power	Failingtodetectarealeffect;power=$1-\beta$.30replicationspergroupgiveabout 80% power for effect sizes d $\geq$0.75 (see ALA 3).
Effect size	Cohen's d (difference in means divided by the pooled SD) and the 95% confidence intervalofthedifference:theyshow*howlarge*aneffectis,notonlywhetheritexists.
Unit of analysis	Onereplication=oneindependentrandomdataset(seed).Thedependentvariable of a replication is the mean number of nodes inspected per search over 3000 queries.

3. Variables

Role	Variable	Levels / measurement
Independent	Data structure	Skip list (p = 0.5), red-black tree, splay tree (nominal)
Independent	Insertion order	Random, sorted (nominal)
Independent	Query pattern	Uniform; skewed (90% of queries to 1% of the keys) (nominal)
Dependent	Search cost	Mean nodes inspected per successful search (ratio scale)
Controlled	Size and implementation	n=5000keys,3000queriesperrun,samelanguage,machine and key range 1–999,999

4. Formulation of Hypotheses from the Literature

Each hypothesis is derived from a finding or gap identified in the literature review.

ID	Source in review	Null hypothesis H_0	Alternative H_1	Test
H1	P2, P5, P7: skip list expected cost $\approx(\log_2 n)/p$, red-black height $\approx\log_2 n$	Meansearchcostofskiplist = red-black tree (uniform queries,randominsertion)	Meansdiffer (two-tailed)	Welch t-test
H2	P3:splaytreesadaptto skewed access	Meansearchcostofsplay= red-black tree (skewed queries)	Splayislower (one-tailed)	Welch t-test, one-tailed
H3	P4, P5: randomized structuresdonotdepend on insertion order	Skip-listcostisthesamefor sorted and random insertion	Costsdiffer (two-tailed)	Welcht-test+equivalence test (TOST, ±1 node)
H4	P3: splay-tree shape dependsontheaccess history	Splay-treecostisthesame for sorted and random insertion	Costsdiffer (two-tailed)	Welch t-test
H5	P5,P7:nodeheightsare geometric with p = 0.5	Levelfrequenciesfollow Geometric(0.5)	They do not	Chi-square goodness of fit
H6	CombinedviewofP2,P3, P5	All three structures have equalmeancost(uniform queries)	Atleastonemean differs	One-wayANOVA(+ Tukey HSD)

5. Selection of Statistical Techniques

Situation	Technique	Used here?
Comparemeansof2independent groups, unequal variances	Welch's t-test (does not assume equal variances)	Yes: H1–H4
Same,butdatanotnormal/small samples	Mann–Whitney U test (rank-based)	Yes:robustness check
Compare means of 3 or more groups	One-way ANOVA; post-hoc Tukey HSD; Kruskal–Wallis as non-parametric version	Yes: H6
Observedvsexpectedfrequenciesof categories	Chi-square goodness-of-fit test	Yes: H5
Paired measurements on the same units	Paired t-test / Wilcoxon signed-rank	No(groupsare independent)
Association of two numerical variables	Pearson / Spearman correlation, regression	Not needed
Many tests at once	Holm (or Bonferroni) correction of p-values	Yes: H1–H4

6. Checking the Assumptions

Normality of the 30 replication means was checked with the Shapiro–Wilk test and equality of variances with Levene's test (uniform queries, random insertion).

Group	Shapiro–Wilk W	p-value
Skip list (random, uniform)	0.951	0.184
Red-black (random, uniform)	0.979	0.791
Splay (random, uniform)	0.934	0.063

Nogroupdepartssignificantlyfromnormalityatα=0.05(allp>0.05).Levene'stestgivesW=49.963,p< 0.001: the variances are clearly **unequal** (the red-black tree is almost deterministic, the skip list is random). This is why Welch's t-test (which does not need equal variances) was used instead of the ordinary t-test, and why the Mann–Whitney test is reported as a check.

7. Descriptive Results

Structure	Insertion	Queries	n	Mean	SD	95% CI of mean	µs/search
SkipList	random	uniform	30	24.31	1.69	[23.68, 24.95]	1.98
SkipList	random	skewed	30	24.38	1.9	[23.67, 25.09]	1.98
SkipList	sorted	uniform	30	24.73	1.8	[24.06, 25.41]	2.17
SkipList	sorted	skewed	30	24.78	1.86	[24.08, 25.47]	1.89
RedBlack	random	uniform	30	11.62	0.04	[11.61, 11.64]	0.77
RedBlack	random	skewed	30	11.59	0.23	[11.51, 11.68]	0.67
RedBlack	sorted	uniform	30	11.89	0.04	[11.88, 11.91]	0.75
RedBlack	sorted	skewed	30	11.89	0.24	[11.80, 11.98]	0.67
Splay	random	uniform	30	21.26	0.09	[21.22, 21.29]	2.45
Splay	random	skewed	30	10.33	0.09	[10.30, 10.36]	1.31
Splay	sorted	uniform	30	23.59	0.11	[23.55, 23.63]	2.73
Splay	sorted	skewed	30	12.87	0.11	[12.83, 12.91]	1.61

Mean=average(over30replications)ofnodesinspectedpersearch;µs/search=Pythonwall-clocktime(machine dependent, secondary measure).

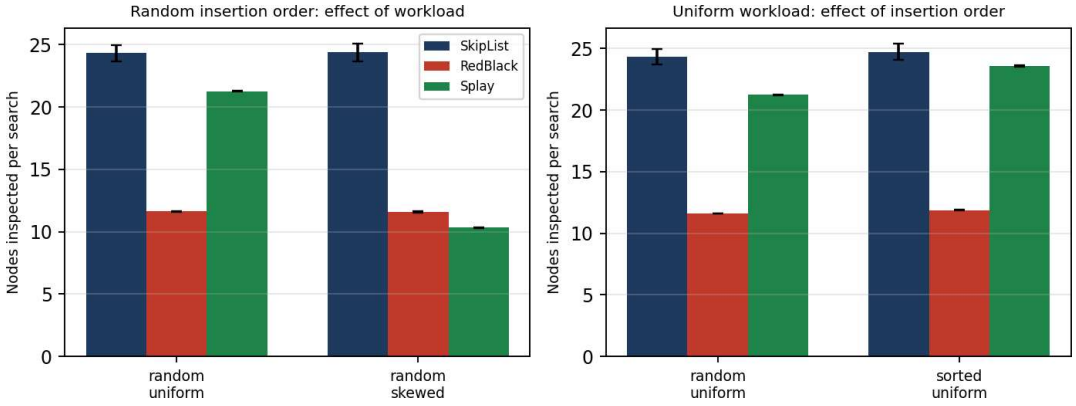


Figure 1: Mean search cost with 95% confidence intervals.

8. Hypothesis Test Results

ID	Comparison	Mean diff.	95% CI of diff.	t	df	p	Holm p	MW p	d
H1	Skip list – Red-black(uniform, random)	12.69	[12.06, 13.32]	41.15	29.0	<0.001	<0.001	<0.001	10.62
H2	Splay–Red-black (skewed, random)	-1.26	[-1.35, -1.17]	-28.02	37.0	<0.001	<0.001	<0.001	-7.23
H3	Skip list: sorted – random (uniform)	0.42	[-0.48, 1.32]	0.93	57.8	0.358	0.358	0.559	0.24
H4	Splay: sorted – random (uniform)	2.34	[2.29, 2.39]	94.06	55.6	<0.001	<0.001	<0.001	24.29

*H2isone-tailed(splaylower),soitsCIistwo-sidedanditsp-valueisone-sided.HolmcorrectionwasappliedtothefourH

H5: chi-square test of the skip-list level distribution

50,000 random levels were generated by the skip list's level generator (p = 0.5). Expected counts underGeometric(0.5) are $N \cdot (1/2)^h$; the last categories were merged so that every expected count is at least 5.

Level h	Observed	Expected
1	24977	25000.0
2	12415	12500.0
3	6251	6250.0
4	3201	3125.0
5	1580	1562.5
6	785	781.2
7	421	390.6
8	178	195.3
9	94	97.7
10	47	48.8
>=11	51	48.8

χ^2=6.86,df=10,p=0.739.Sincep>0.05,H $_0$ is**notrejected**:theobservedlevelfrequenciesare consistent with the geometric distribution.

9. Interpretation of Results

Hypothesis	Decision	Interpretation
H1	Reject H_0	The skip list inspects about 12.69 more nodes per search than the red-black tree (24.31 vs 11.62; about 2.1 times). The resultagrees with theory: $2 \cdot \log_2(5000) \approx 24.6$ for the skip list and $\log_2(5000) \approx 12.3$ for abalanced tree. The effect is very large (d > 10) because the red-black cost barely varies between replications.
H2	Reject H_0	Withskewedqueriesthesplaytreeneeds10.33nodespersearchagainst11.59 for the red-black tree, about 1.26 fewer (about 11%). Adaptation helps when a few keys are queried most often.
H3	Fail to reject H_0	No significant difference (24.73 vs 24.31; p = 0.358), consistent with order-independence, but equivalence within ±1 node could not be demonstrated.
H4	Reject H_0	Sortedinsertionleavesthesplaytreeinapoorershape:23.59vs21.26nodes per search (+2.34); the structure depends on history.
H5	Fail to reject H_0	Therandomlevelsfollowtheintendedgeometricdistribution(thegenerator behaves as designed).
H6	Reject H_0	Thethreestructuresdifferstronglyinmeancost;allthreepairsdiffer(Tukey HSD).

10. Limitations of the Testing

- Statistical significance is easy to obtain here because the number of replications is a free choice and the variance of some groups is tiny; the **effect sizes** and confidence intervals are therefore more informative than p-values.
- The results describe the *implementations used* (Python, nodes inspected as the metric, n = 5000). Counting rules differ slightly between structures (see the Research Report, threats to validity).
- Four primary hypotheses were tested; Holm correction was applied and does not change any decision.

11. Conclusion

Six hypotheses were formulated from the literature and tested with techniques matched to the data: Welch's t-test for pairwise comparisons, one-way ANOVA with post-hoc test for the three-group comparison, a chi-square test for the distributional hypothesis and an equivalence test for the ‘no difference’ claim. The tests support the predictions that the red-black tree has the lowest search cost for uniform queries, that the splay tree wins under skewed access, and that insertion order matters for the splay tree but not detectably for the skip list.

References

- C. R. Kothari and G. Garg, *Research Methodology: Methods and Techniques*, New Age International Publishers (hypothesis testing).
- R. Kumar, *Research Methodology: A Step-by-Step Guide for Beginners*, SAGE Publications.
- J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed., Lawrence Erlbaum, 1988.
- A. Georges, D. Buytaert, L. Eeckhout, “Statistically rigorous Java performance evaluation”, *OOPSLA'07*, 2007.
- Literature-review references P3, P5, P7 as listed in ALA 1.